

Exercises from Lectures 1–3. Throughout, $\mathbb{k}$ is a field, and all modules are right modules unless a left action is specified. Write $\mathbb{k}A_n$ for the algebra of upper triangular $n \times n$ matrices over $\mathbb{k}$, with matrix units e_{ij} ($1 \leq i \leq j \leq n$) and grading $\deg e_{ij} = j - i$. For a graded module M , the shift convention is $M(r)_d = M_{d+r}$. Ideals in algebra presentations are two-sided ideals.

Lecture 1: Graded algebras, modules, and duals

Exercise 1. Graded modules and homogeneous maps

The following parts are independent.

- (1) Let $n \geq 1$ and $A = \mathbb{k}A_n$. Verify that

$$A_d = \text{span}_{\mathbb{k}}\{e_{ij} \mid j - i = d\}$$

is a grading, using $e_{ij}e_{pq} = \delta_{jp}e_{iq}$. Compute $\text{rad}(A)$ from its definition as the intersection of all maximal right ideals.

- (2) Let $A = \mathbb{k}A_3$ and $M = e_{22}A$. Its graded linear dual is

$$\mathbb{D}(M)_d = \text{Hom}_{\mathbb{k}}(M_{-d}, \mathbb{k}), \quad (af)(m) = f(ma) \quad (a \in A, f \in \mathbb{D}(M), m \in M).$$

Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of the matrix units on that basis. Show that $\mathbb{D}(M)$ is not isomorphic to any left module $(Ae_{jj})(r)$, for $j \in \{1, 2, 3\}$ and $r \in \mathbb{Z}$.

Hint. Compare dimensions and the actions of e_{11}, e_{22}, e_{33} .

- (3) Let $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$, with $e_i e_j = \delta_{ij} e_i$ and $1 = e_1 + e_2$, in degree 0. Set

$$M = \bigoplus_{i \geq 0} \mathbb{k}m_i, \quad \deg m_i = i, \quad m_i e_1 = m_i, \quad m_i e_2 = 0,$$

and let $N = \mathbb{k}e_1$ be in degree 0. Define the S -linear map $f : M \rightarrow N$ by $f(m_i) = e_1$ for every $i \geq 0$. Write every non-zero homogeneous component $f_d : M \rightarrow N$ explicitly, where a degree d map sends M_j into N_{j+d} . Prove that f is not a finite sum of homogeneous maps.

- (4) Let $A = \mathbb{k}A_3$ and $P_i = e_{ii}A$ for $i \in \{1, 2, 3\}$, with the inherited grading.
- For each of the nine ordered pairs (i, j) , find a basis of $\text{Hom}_A(P_i, P_j)$ and the degree of every basis map. Present the answers in a 3×3 table whose rows index the source P_i and whose columns index the target P_j .
 - Compute the composite

$$P_3 \xrightarrow{p \mapsto e_{23}p} P_2 \xrightarrow{p \mapsto e_{12}p} P_1.$$

Determine the degree of each map and of the composite.

- For every $i, j \in \{1, 2, 3\}$ and $r, s \in \mathbb{Z}$, determine when there is a non-zero degree-preserving map $P_i(r) \rightarrow P_j(s)$.

Hint. An A -linear map $u : P_i \rightarrow P_j$ is determined by $u(e_{ii})$, and $u(e_{ii}) = u(e_{ii})e_{ii}$.

Exercise 2. Dependence on the symmetrising form

Let $A = \mathbb{k}A_2$, $S = \mathbb{k}e_{11} \oplus \mathbb{k}e_{22}$, $V = \mathbb{k}e_{12}$, and $M = e_{11}A$, with their multiplication actions and inherited gradings. For $c_1, c_2 \in \mathbb{k} \setminus \{0\}$, define

$$\tau : S \longrightarrow \mathbb{k}, \quad ae_{11} + be_{22} \longmapsto c_1a + c_2b.$$

Recall that $M^{(*)} = \bigoplus_{d \in \mathbb{Z}} \text{Hom}_S(M_{-d}, S)$, with the displayed summand in degree d . A form is symmetrising when $\tau(st) = \tau(ts)$ for all $s, t \in S$ and the pairing $(s, t) \mapsto \tau(st)$ is non-degenerate. For parts (3)–(4), take $\mathbb{k} = \mathbb{Q}$.

- (1) Verify that τ is symmetrising. Define $\varepsilon : V \rightarrow \mathbb{k}$ by $\varepsilon(ue_{12}) = u$ for $u \in \mathbb{k}$. Find the right S -linear map $f : V \rightarrow S$ and the left S -linear map $g : V \rightarrow S$ satisfying $\tau \circ f = \varepsilon = \tau \circ g$. Verify the stated linearities and determine whether f is also left S -linear and whether g is also right S -linear.
Check. $f(e_{12}) = c_2^{-1}e_{22}$ and $g(e_{12}) = c_1^{-1}e_{11}$.
- (2) Consider the right S -linear maps

$$\begin{aligned} \varphi_0 : M &\longrightarrow S, & ae_{11} + be_{12} &\longmapsto ae_{11}, \\ \varphi_1 : M &\longrightarrow S, & ae_{11} + be_{12} &\longmapsto be_{22} \end{aligned} \quad (a, b \in \mathbb{k}).$$

Verify that φ_0, φ_1 form a homogeneous basis of $M^{(*)}$, and determine their degrees.

- (3) Set $\tau_1(ae_{11} + be_{22}) = a + b$ and $\tau_2(ae_{11} + be_{22}) = a + 2b$. For $i \in \{1, 2\}$, define

$$\alpha_i : M^{(*)} \longrightarrow \mathbb{D}(M), \quad f \longmapsto \tau_i \circ f, \quad a \cdot_i f = \alpha_i^{-1}(a\alpha_i(f)),$$

where $(a\lambda)(m) = \lambda(ma)$ for $a \in A$, $\lambda \in \mathbb{D}(M)$, and $m \in M$. Calculate the actions of e_{11}, e_{22}, e_{12} on φ_0, φ_1 for both forms. Are the two actions equal?

- (4) Construct a graded left A -module isomorphism

$$t : (M^{(*)}, \cdot_1) \longrightarrow (M^{(*)}, \cdot_2)$$

by specifying $t(\varphi_0)$ and $t(\varphi_1)$.

Lecture 2: Tensor and quadratic algebras

Exercise 3. Interaction of tensor product with S -dual

Let S be a finite-dimensional semisimple $\mathbb{k}$ -algebra, and let V, W be finite-dimensional S -bimodules. For any such bimodule U , let

$$U^* = \text{Hom}_S(U, S)$$

be the space of right S -linear maps, with bimodule action

$$(sft)(u) = sf(tu) \quad (s, t \in S, f \in U^*, u \in U).$$

For $f \in V^*, g \in W^*, v \in V$, and $w \in W$, consider

$$\begin{aligned} \theta_r^{V,W} : W^* \otimes_S V^* &\longrightarrow (V \otimes_S W)^*, \\ \theta_r^{V,W}(g \otimes f)(v \otimes w) &= g(f(v)w). \end{aligned}$$

Prove that $\theta_r^{V,W}$ is a natural isomorphism of S -bimodules.

- (1) Prove that the formula defines an S -bimodule map. Explicitly, check bilinearity in (g, f) and in (v, w) , and show that neither of the substitutions

$$vs \otimes w \longleftrightarrow v \otimes sw, \quad gs \otimes f \longleftrightarrow g \otimes sf \quad (s \in S)$$

changes the value given by the formula. Check that the resulting map $V \otimes_S W \rightarrow S$ is right S -linear, and verify

$$\theta_r^{V,W}(sg \otimes ft) = s\theta_r^{V,W}(g \otimes f)t \quad (s, t \in S).$$

- (2) Prove that $\theta_r^{V,W}$ is bijective.
 (3) For finite-dimensional S -bimodules V', W' and bimodule maps $p : V \rightarrow V'$ and $q : W \rightarrow W'$, prove

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

Here $p^* : (V')^* \rightarrow V^*$ sends f to $f \circ p$; $q^* : (W')^* \rightarrow W^*$ and $(p \otimes q)^* : (V' \otimes_S W')^* \rightarrow (V \otimes_S W)^*$ are defined in the same way by composition with q and $p \otimes q$, respectively.

Hint for (2). First take $V = S$. Then use the fact that every finite-dimensional right S -module is a direct summand of $S^{\oplus n}$ for some $n \geq 0$.

Remark. The analogous statement holds for left S -duals, with left and right interchanged.

Exercise 4. Quadratic presentations and Hilbert series

The following parts are independent. Throughout this sheet, for an algebra $A = \bigoplus_{d \geq 0} A_d$ with $A_0 = \bigoplus_{i=1}^r \mathbb{k}e_i$, use

$$H_A(t)_{ij} = \sum_{d \geq 0} \dim_{\mathbb{k}}(e_i A_d e_j) t^d \quad (1 \leq i, j \leq r),$$

with the idempotents in their listed order. Thus the coefficient of t^d is the dimension vector of the A_0 -bimodule A_d , recorded with two indices as in Lecture 2. When $A_0 = \mathbb{k}$, there is just one entry.

- (1) Let $A = \mathbb{k}A_4/(e_{13}, e_{24})$ with the quotient grading. Find a homogeneous basis, compute $H_A(t)$, and write $A = T_S(V)/(R)$ with explicit S , V , and $R \subset V \otimes_S V$.
- (2) Let $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$ with coordinate idempotents, and let $V = \mathbb{k}a \oplus \mathbb{k}b$ satisfy

$$e_1 a e_2 = a, \quad e_2 b e_1 = b,$$

with all other idempotent actions zero. Place V in degree 1. Find a basis of $T_S(V)_d$ for every $d \geq 0$, and show that

$$H_{T_S(V)}(t) = \frac{1}{1-t^2} \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

Find a homogeneous basis and the Hilbert series of $T_S(V)/(ab)$, and compare the two algebras.

Hint. Count alternating words separately in each degree, including degree 0.

- (3) Let V and W have bases $x_1, \dots, x_n$ and $y_1, \dots, y_m$, respectively. Give every generator degree 1 and set

$$\begin{aligned} \text{Sym}(V) &= T_{\mathbb{k}}(V)/(x_i x_j - x_j x_i \mid 1 \leq i < j \leq n), \\ \bigwedge W &= T_{\mathbb{k}}(W)/(w^2 \mid w \in W). \end{aligned}$$

Prove that

$$\dim_{\mathbb{k}} \text{Sym}(V)_2 = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{k}} (\bigwedge W)_2 = \frac{m(m-1)}{2}.$$

Lecture 3: The quadratic dual

For $A = T_S(V)/(R)$, use the right quadratic dual $A^! = T_S(V^*)/(R^\perp)$, with $V^* = \text{Hom}_S(V, S)$ placed in degree 1. The reversed pairing is

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w).$$

When $S = \bigoplus_i \mathbb{k}e_i$ and $v \in e_i V e_j$ is a chosen basis vector, its dual $v^* : V \rightarrow S$ takes v to e_j and the other basis vectors to zero; it satisfies $e_j v^* e_i = v^*$.

Exercise 5. Computing quadratic duals

The following parts are independent.

- (1) **Reverse generators and complement relations.** Let $A = \mathbb{k}A_4/(e_{13})$. Write a presentation of $A^!$ using reversed generators, and find homogeneous bases of A and $A^!$. Check directly that dualising twice recovers A , identifying each twice reversed generator with its original matrix unit.
Hint. Among $e_{12}e_{23}$ and $e_{23}e_{34}$, only the first is a relation. Complement the forbidden composable pairs and reverse their order.
- (2) **All products of two generators vanish.** Let $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$ and $V = \mathbb{k}a \oplus \mathbb{k}b$, with $e_1 a e_2 = a$ and $e_2 b e_1 = b$ and all other idempotent actions zero. Compute the quadratic dual of $T_S(V)/(ab, ba)$ and the Hilbert series of both algebras.
Check. In the order e_1, e_2 , the two series are

$$\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix} \quad \text{and} \quad \frac{1}{1-t^2} \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

- (3) **When does every linear combination square to zero?** Write $\mathbb{k}\langle x, y \rangle := T_{\mathbb{k}}(\mathbb{k}x \oplus \mathbb{k}y)$; its basis consists of the finite words in x, y , including the empty word 1, and multiplication joins words in order. For $q \in \mathbb{k}$, let

$$C_q = \mathbb{k}\langle x, y \rangle / (x^2, y^2, xy + qyx), \quad \deg x = \deg y = 1.$$

- i. Find a homogeneous basis of C_q , compute every product of two basis elements, and determine $H_{C_q}(t)$. Treat $q = 0$ as well as $q \neq 0$, and justify that your proposed basis is linearly independent.
- ii. For $u = \alpha x + \beta y \in (C_q)_1$, compute u^2 . Determine, for each q , exactly which pairs $(\alpha, \beta) \in \mathbb{k}^2$ give $u^2 = 0$. For which q does every element of $(C_q)_1$ square to zero?
- iii. Compute a quadratic presentation of $C_q^!$ using the reversed pairing. Identify C_1 and C_{-1} as the graded and ordinary tensor products, respectively, of two copies of $\mathbb{k}[z]/(z^2)$ with $\deg z = 1$. When the characteristic of $\mathbb{k}$ is not 2, are C_1 and C_{-1} isomorphic as algebras? Explain what changes in characteristic 2.

Hint. In a candidate basis, use the class of yx for degree 2. For the dual relation, evaluate a general linear combination of X^2, XY, YX, Y^2 on $x^2, y^2, xy + qyx$, where $X = x^*$ and $Y = y^*$.

Exercise 6. Bases, Hilbert series, and quadratic duals

The following parts are independent.

- (1) **A square with a parameter.** Let $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2 \oplus \mathbb{k}e_3 \oplus \mathbb{k}e_4$ and let V have basis a, b, c, d , with

$$e_1ae_2 = a, \quad e_2be_4 = b, \quad e_1ce_3 = c, \quad e_3de_4 = d,$$

and all other idempotent actions zero. The only composable words of length 2 are ab and cd . For each $\lambda \in \mathbb{k}$, set

$$A_\lambda = T_S(V)/(ab - \lambda cd).$$

For every $\lambda \in \mathbb{k}$, including $\lambda = 0$, compute a presentation of $A_\lambda^!$. Find homogeneous bases of A_λ and $A_\lambda^!$, and compute both Hilbert series. Do the series depend on λ ? Recover the original relation by dualising again. For $\lambda \neq 0$, rescale d to compare A_λ with A_1 .

Hint. Evaluate $\mu b^* a^* + \nu d^* c^*$ on $ab - \lambda cd$, where $\mu, \nu \in \mathbb{k}$.

- (2) **Counting words.** Let $V = \mathbb{k}x \oplus \mathbb{k}y$ and

$$B = T_{\mathbb{k}}(V)/(x^2), \quad \deg x = \deg y = 1.$$

- i. Describe a basis of B_d for every $d \geq 0$, and compute $a_d := \dim_{\mathbb{k}} B_d$ for $0 \leq d \leq 6$.
- ii. For $d \geq 2$, find an equation expressing a_d in terms of a_{d-1} and a_{d-2} by sorting the basis words according to their first letter. Use the equation and the values of a_0, a_1 to find $H_B(t)$ as a rational function.
- iii. Write $X = x^*$ and $Y = y^*$. Compute $B^!$, find a homogeneous basis, and determine $H_{B^!}(t)$. Which of B and $B^!$ are finite-dimensional?